

Vishwas Prajapati

Jhalawar (Rajasthan) | (+91) 9216788910 | vishwas1423@gmail.com | [GitHub](#)

OBJECTIVE

Computer Science & Engineering student specializing in embedded systems, seeking hands-on opportunities to apply and expand my skills through real-world projects and industry exposure.

EDUCATION

Government Engineering College, Jhalawar

September 2024 – Present

Bachelor of Technology in Computer Science & Engineering

8.34 CGPA

PROJECTS

STM32 Bare-Metal Driver Development

- Developed bare-metal peripheral drivers in C for the STM32 Nucleo-F446RE, including GPIO, DMA, NVIC/Interrupts, ADC, Timers, PWM, DAC, UART, I2C, SPI, and CAN.
- Gained a deep understanding of STM32 peripheral architecture, ARM Cortex-M4 microcontroller operation, interrupt handling, and register-level programming.
- Worked extensively with STM32 reference manuals, datasheets, and technical documentation to design, implement, and debug firmware.
- Applied embedded C concepts such as bit manipulation, structures, volatile qualifiers, and memory-mapped I/O to build reusable and modular drivers.

Automated Solar Panel Cleaner

- Designed and developed an autonomous solar panel cleaning prototype to improve panel efficiency by removing dust and debris.
- Built the mechanical and embedded control system using an STM32 microcontroller, stepper motors, motor drivers, and cleaning mechanisms.
- Implemented firmware in Embedded C to control motor movement, cleaning sequences, and peripheral interfaces.
- Integrated hardware components including sensors, actuators, and power electronics to automate the cleaning process.

SKILLS

Programming Languages: C, C++, Python

Development Tools & IDEs: Git, STM32CubeIDE, Keil μ Vision, VS Code

Microcontrollers: STM32 (Cortex-M series)

Communication Protocols: UART, SPI, I2C

Core Concepts: Digital Electronics, ARM Cortex-M4 Architecture, Operating Systems, Object Oriented Programming, Computer Organization and Architecture

CERTIFICATES

Udemy's Foundation of Embedded Systems with ARM Cortex and STM32

Gained in-depth knowledge of ARM Cortex-M fundamentals, including registers, exceptions, privilege levels, and endianness. Set up an open-source toolchain for programming, flashing, and debugging using OpenOCD and STM32CubeMX. Learned peripheral programming, HAL libraries, vector tables, VTOR, and built confidence in Cortex-M application development.